

Software Test Plan (STP)

Event Management System

Version: 1.0

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Project: A web-based system to manage and organize events with participant registrations.

Technology: React.js frontend, Node.js/Express.js backend, MongoDB database, REST APIs with JSON.

Primary users: Participant, Organizer and Administrator.

1. Introduction

Purpose: This Software Test Plan defines the testing objectives, scope, strategy, environment, responsibilities, risks, traceability and reporting approach for the Event Management System (EMS) Version 1.0.

Scope: Testing covers user registration and authentication, role-based authorization, event creation/editing/publishing/cancellation, event browsing/search/filtering, event details, participant registration/cancellation/history, capacity and deadline enforcement, organizer registration management, administrator user/event management, confirmations/notifications, audit records, performance, usability, data integrity and security.

Out of scope: Version 1.0 excludes payment processing, specialized QR/barcode hardware, external calendar synchronization and production email/SMS infrastructure. The system has no dedicated physical hardware interface.

References: Event Management System SRS v1.0 dated 12-09-2026; supplied Software Test Plan template; the SRS Requirements Traceability Matrix and acceptance-test references.

Definitions: EMS – Event Management System; FR – Functional Requirement; NFR – Non-Functional Requirement; SRS – Software Requirements Specification; RTM – Requirements Traceability Matrix; RBAC – Role-Based Access Control; UAT – User Acceptance Testing; STP – Software Test Plan.

2. Test Items

- User Registration & Authentication module.
- Role-Based Authorization and protected-access controls.
- Event Creation, Editing, Publishing and Cancellation module.
- Event Discovery, Search, Filtering and Event Details.
- Participant Registration, Cancellation and Registration History.
- Capacity, registration-deadline and duplicate-registration enforcement.
- Organizer Registration Count and Participant List management.
- Administrator User and Event Management.
- Notifications / confirmation feedback and Audit Records.
- Performance, usability, data-integrity and security controls.
- React.js frontend, Node.js/Express.js REST API layer and MongoDB persistence integration.

3. Features to be Tested

Functional coverage: The SRS defines 27 functional requirements (EMS-F-001 through EMS-F-027). The test plan covers the complete functional scope through the following test suites and mappings.

Feature / SRS area	SRS IDs	Primary test cases
Registration & Authentication	EMS-F-001–004	TC-AUTH-01, TC-AUTH-02
Event Creation / Editing / Publishing / Cancellation	EMS-F-005–008	TC-EVENT-01 to TC-EVENT-06
Event Discovery / Search / Filter / Details	EMS-F-009–012	TC-EVENT-05, TC-EVENT-06
Registration & Cancellation	EMS-F-013–017	TC-REG-01 to TC-REG-06
Organizer Registration Management	EMS-F-018–020	TC-ORG-01
Administration	EMS-F-021–023	TC-ADMIN-01, TC-AUTHZ-02
Notifications & Audit	EMS-F-024–027	TC-AUDIT-01 and notification checks
Performance	EMS-NF-001–003	TC-PERF-01 to TC-PERF-03
Data Integrity	EMS-NF-004	TC-DATA-01
Usability / Browser Viewports	EMS-NF-005	TC-UX-01
Logging / Auditability	EMS-NF-006	TC-OPS-01
Security Requirements	EMS-SR-001–006	TC-SEC-01 to TC-SEC-03, TC-AUTHZ-01/02, TC-AUDIT-01

4. Features Not to be Tested

- Payment processing and external payment gateway integration.
- Specialized QR/barcode hardware and physical event-management devices.
- External calendar synchronization.
- Production email/SMS infrastructure; Version 1.0 uses in-app notifications.
- Physical safety-critical functions, because the academic web application does not control physical equipment.
- Internal implementation of external services that are outside the EMS boundary.

5. Test Approach / Strategy

5.1 Test Levels

- Unit Testing – test individual frontend/backend modules and business-rule functions in isolation where applicable.
- Integration Testing – verify React.js ↔ REST API ↔ MongoDB interactions and authorization/business-rule integration.
- System Testing – verify complete Participant, Organizer and Administrator workflows end-to-end.
- Acceptance Testing (UAT) – verify that the implemented system satisfies the SRS acceptance criteria and demonstration workflows.

5.2 Test Types

- Functional Testing – verify all required user and administrative functions.
- Negative / Boundary Testing – invalid credentials, duplicate registrations, full events, expired deadlines and unauthorized operations.
- Regression Testing – re-run affected tests after defect fixes or changes.
- Performance Testing – verify the SRS response-time targets under normal demonstration load.

- Usability Testing – verify core workflows on representative desktop and mobile browser viewports.
- Security Testing – verify password handling, authorization, input validation, safe responses/logs and security auditing.
- Data Integrity / Concurrency Testing – verify no duplicate active registration and no over-capacity registration.

5.3 Entry Criteria

- A stable build is available for testing.
- Frontend, backend and MongoDB services can be started successfully.
- Required test accounts and representative event data are available.
- Core authentication and database connectivity are functional enough to execute planned tests.
- Test environment and browser access are ready.

5.4 Exit Criteria

- 100% of planned test cases have been executed or formally deferred with justification.
- No unresolved critical security or data-integrity defect remains.
- High-priority functional requirements have been verified.
- Required acceptance test suites have been executed.
- Performance, authorization, registration-integrity and security acceptance criteria are satisfied or formally recorded as limitations.
- RTM contains traceability from requirements to test cases and execution status.

5.5 Security Validation

Security area	Validation
Password protection	Verify passwords are stored only as salted one-way hashes and never as plaintext. TC-SEC-01.
Server-side authorization	Attempt protected Organizer/Administrator API operations with insufficient privileges. TC-AUTHZ-01/02.
Input validation	Submit malformed/injection-style values and verify safe rejection. TC-SEC-02.
Secrets / error disclosure	Inspect normal responses and logs to ensure passwords, tokens, secrets and internal stack traces are not exposed. TC-SEC-03.
Security audit	Verify repeated authentication failures and privilege changes are timestamped in audit records. TC-AUDIT-01.

6. Test Environment

Application stack: React.js frontend, Node.js/Express.js backend and MongoDB persistence, communicating through REST APIs using JSON.

Execution environment: Academic/local demonstration environment using team computers and a modern web browser such as Chrome. The SRS also permits deployed HTTPS for environments handling real credentials.

Hardware: Standard laptops/desktops/tablets/smartphones with supported browsers. No specialized hardware is required.

Network: Normal network connectivity for client-server operation; deployed environments handling real credentials should use HTTPS.

Suggested tools: Browser developer tools for UI/network inspection, Postman or an equivalent REST client for API verification, and the team's shared version-control repository for test artifacts and defect-fix tracking. Tool choice may be adjusted by the team.

Test data: Representative Participant, Organizer and Administrator accounts; valid/invalid credentials; published and unpublished events; events with different capacities and deadlines; registrations, cancellations and duplicate-registration attempts; unauthorized-role requests; malformed input values; and audit/security events.

Normal demonstration load: A controlled local/academic workload representative of the normal project demonstration, rather than a stress test. For each performance execution, the actual sample size/concurrency and environment conditions shall be recorded so that the 2–5 second SRS targets have a defined test condition.

7. Test Schedule

Note: The following is a proposed project schedule created for planning. Actual execution dates may be adjusted by the team.

Milestone	Proposed timing	Output
Test plan finalization	28-Sep-2026	Approved STP
Test data & environment setup	29-Sep-2026	Ready test environment
Unit / API / module testing	30-Sep 2026	Unit/API results
Integration & system testing	30–31 Sep 2026	System test results
Security, performance & usability	01-Oct-2026	NFR/security results
UAT / demonstration validation	02-Oct-2026	Acceptance results
Defect re-test & regression	03-Oct-2026	Regression results
Test summary / sign-off	04-Oct-2026	Final Test Summary Report

8. Test Deliverables

- Software Test Plan (this document).
- Functional and non-functional test cases.
- API / integration test records.
- Test data and test-account details.
- Test execution logs and screenshots where useful.
- Defect reports with severity, steps, expected result and actual result.
- Regression-test results.
- Security, performance, usability and data-integrity test evidence.
- Updated Requirements Traceability Matrix.
- Final Test Summary Report.

9. Roles and Responsibilities

Team member / role	Primary test responsibility
ARYA V D	Organizer registration-management and participant-list testing; Administrator user/event-management testing; notifications/audit testing; authorization and auditability checks.
PRANEETH P SHETTY	Participant registration/cancellation/history testing; capacity, deadline and duplicate-registration testing; data-integrity, edge-case and API integration testing.
SINDHU GV	User registration/login/authentication testing; participant-profile and role-based authorization testing; credential-handling, security and usability checks.
PREETHI T	Event creation/edit/publish/cancel testing; browsing/search/filter/event-details testing; performance, API-efficiency, maintainability and UI integration checks.
Project Team — Shared	Environment setup, regression testing, defect triage, code review support, database integration, final integration and test-summary preparation.
Course Coordinator	Review/approval of the project and acceptance readiness as applicable to the academic evaluation.

Role	Name	Responsibility
QA Lead	Preethi T	Prepare plan, coordinate execution
Test Engineer	Sindhu G V	Design & execute test cases, log defects
Developer	Arya V D	Support defect fixes and triage
Product Owner	Praneeth P Shetty	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Impact	Mitigation
Backend/database environment unavailable	Blocks most tests	Maintain a reproducible local setup and verify frontend/backend/database startup before execution.
Free-tier / hosted cold-start delay	Can distort performance results	Exclude cold-start requests from performance targets as specified in SRS; run performance tests only after the backend is warm and record the condition.
Frontend/backend API mismatch	Integration failures	Use API checks and integration tests; validate request/response contracts before full system testing.
Capacity / duplicate-registration race condition	Data-integrity failure	Perform edge/concurrency tests around full capacity and repeated registration; verify database state after execution.
Incorrect role authorization	Security/privacy risk	Run explicit Participant/Organizer/Administrator boundary tests against protected API operations.
Unstable build during testing	Test execution blocked	Use smoke testing after each significant build and suspend affected suites until blocking defects are fixed.
Incomplete test data	Coverage gaps	Prepare representative users, events, deadlines, capacities and registrations before execution.

11. Assumptions & Dependencies

- Users have a supported browser and network connection.
- Backend and MongoDB services are available during normal operation.
- The system clock is correctly configured for event dates and registration deadlines.
- The academic demonstration is performed under controlled local/academic load.
- Version 1.0 does not depend on an external email/SMS provider; in-app notifications are sufficient.
- The test build reflects the requirements and module ownership documented in the SRS.
- The team's shared repository and development workflow remain available for test artifacts and fixes.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment or a required core service is unavailable and prevents execution of a substantial portion of the planned tests.
- A build is sufficiently unstable that a significant portion of test cases cannot be meaningfully executed.
- A critical security or data-integrity defect makes continued testing unsafe or invalid.
- Required test data or database state is corrupted and cannot be trusted.

Resume testing if:

- Blocking defects are fixed and a stable build is delivered.
- The environment and database state have been restored and verified.
- Required test data is recreated or validated.
- A smoke test confirms that the affected core workflows are operational.

13. Test Case Management & Traceability

Traceability approach: The SRS RTM maps selected requirements to design/module references and test cases. This STP extends that approach to the complete test scope. Test-case status should be maintained as Not Executed, Passed, Failed, Blocked or Not Applicable during execution.

Requirement area	Requirement IDs	Test case mapping
Authentication	EMS-F-001-004	TC-AUTH-01, TC-AUTH-02
Event Management	EMS-F-005-012	TC-EVENT-01 to TC-EVENT-06
Registration	EMS-F-013-017	TC-REG-01 to TC-REG-06
Organizer Management	EMS-F-018-020	TC-ORG-01
Administration	EMS-F-021-023	TC-ADMIN-01, TC-AUTHZ-02
Notifications / Audit	EMS-F-024-027	TC-AUDIT-01 plus notification checks
Performance	EMS-NF-001-003	TC-PERF-01 to TC-PERF-03
Data Integrity	EMS-NF-004	TC-DATA-01

Usability	EMS-NF-005	TC-UX-01
Logging / Auditability	EMS-NF-006	TC-OPS-01
Security	EMS-SR-001-006	TC-SEC-01 to TC-SEC-03; TC-AUTHZ-01/02; TC-AUDIT-01

13.1 Key Acceptance-Test References

- TC-AUTH-01 / TC-AUTH-02 – registration, login, invalid credentials and protected access.
- TC-EVENT-01 to TC-EVENT-06 – create, edit, publish, cancel, view, search/filter and event-management coverage.
- TC-REG-01 to TC-REG-06 – registration, capacity, deadline, duplicate registration, cancellation and status.
- TC-AUTHZ-01 / TC-AUTHZ-02 – Organizer ownership and Administrator role boundaries.
- TC-SEC-01 to TC-SEC-03 and TC-AUDIT-01 – password handling, validation, safe errors and audit.
- TC-PERF-01 to TC-PERF-03 – listing/registration, search/filter and dashboard response times.
- TC-UX-01 – representative desktop/mobile browser workflows.
- TC-DATA-01 – duplicate/over-capacity edge and concurrency conditions.

14. Test Metrics & Reporting

- % of planned test cases executed.
- % of test cases passed, failed, blocked and deferred.
- Requirement coverage across FRs, NFRs and security requirements.
- Defect count by severity and status.
- Defect density and defect aging, where sufficient execution data exists.
- Regression pass rate after defect fixes.
- Performance results for each defined response-time target.
- Security-test pass/fail results.
- Data-integrity test results for duplicate and capacity conditions.

Reports: Daily/iteration execution status as needed, defect reports during execution, and a Final Test Summary Report containing coverage, pass/fail results, outstanding risks and acceptance status.

15. Approvals

Role	Name	Signature / Date
Project Team Representative	ARYA V D / PRANEETH P SHETTY / SINDHU GV / PREETHI T	
Course Coordinator	Dhijo Kumar Behera	

Appendix A: Test Case Catalogue

Test ID	Purpose
TC-AUTH-01	Valid user registration / authentication and successful protected access.
TC-AUTH-02	Invalid credentials, duplicate account identifiers and authentication error handling.
TC-EVENT-01	Create event with valid required details.
TC-EVENT-02	Edit event before it begins according to defined rules.
TC-EVENT-03	Publish event after required information is complete.
TC-EVENT-04	Cancel event and verify new registrations are prevented.
TC-EVENT-05	Browse/view published events and event details.
TC-EVENT-06	Search and filter events using supported criteria.
TC-REG-01	Register authenticated participant for available event.
TC-REG-02	Prevent duplicate active registration.
TC-REG-03	Prevent registration after deadline or at capacity.
TC-REG-04	Cancel permitted participant registration.
TC-REG-05	Verify registration status and capacity after cancellation.
TC-REG-06	Combined registration boundary/negative scenarios.
TC-ORG-01	Organizer views registration count and authorized participant list.
TC-ADMIN-01	Administrator manages users/events through authorized functions.
TC-AUTHZ-01	Reject unauthorized protected Organizer/Administrator API operations.
TC-AUTHZ-02	Verify Administrator-only functions are inaccessible to other roles.
TC-SEC-01	Verify password hashing and absence of plaintext passwords.
TC-SEC-02	Verify server-side input validation and safe rejection.
TC-SEC-03	Verify secrets/tokens/passwords/internal traces are not exposed.
TC-AUDIT-01	Verify timestamped security/admin audit records.
TC-OPS-01	Verify relevant logs can be retrieved for troubleshooting/audit.
TC-PERF-01	Verify event listing and normal registration response target.
TC-PERF-02	Verify event search/filter response target.
TC-PERF-03	Verify Organizer/Admin dashboard response target.
TC-UX-01	Verify core workflows on representative desktop/mobile viewports.
TC-DATA-01	Verify duplicate/over-capacity data-integrity and edge/concurrency behavior.

Appendix B: Project Ownership Used for Testing

- ARYA V D – Organizer registration management, participant lists, Administrator management, notifications and audit records; emphasis on authorization/auditability.
- PRANEETH P SHETTY – registration, cancellation, history, capacity, deadline and duplicate prevention; emphasis on reliability/data integrity.

- SINDHU GV -user registration, login/authentication, participant profiles and RBAC; emphasis on security/usability.
- PREETHI T - event creation/edit/publish/cancel, browsing, search, filter and event details; emphasis on performance/maintainability.